

ZEKI FURKAN YILDIZ

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Education

Ankara Yıldırım Beyazıt University

B.Sc. in Computer Engineering | GPA: 3.32/4.00

Sep. 2021 – Jun. 2026

Ankara, Turkey

Experience

Bilgi Teknolojileri ve İletişim Kurumu (BTK)

Feb. 2026 – May 2026

Long-Term Intern

Ankara, Turkey

- Contributed to an AI Companion mobile application using React Native/Expo, Spring Boot, PostgreSQL, OpenRouter, and FastAPI; developed finance, goals, reminders, and health modules and integrated RESTful APIs with validation, loading states, and asynchronous AI responses.
- Built an EfficientNet-V2-S food-recognition model spanning ~150 Turkish food categories, a K-Means/SVD-based recommendation engine, and a TF-IDF + Logistic Regression content-moderation filter; integrated HealthKit and Google Fit for health-data synchronization.
- Independently designed and developed the Integrated Education Management System (IEMS), a full-stack platform using Spring Boot, PostgreSQL/Supabase, and a Vite-based frontend, with RBAC across five user roles, class/student administration, secure document exchange, lesson scheduling, and exam-calendar generation.

Bilgi Teknolojileri ve İletişim Kurumu (BTK)

Jul. 2025 – Aug. 2025

Intern

Ankara, Turkey

- Developed and maintained a Spring Boot RESTful backend for a restaurant management system using Controller, Service, DTO, Model, and Repository layers.
- Extended stock management with CRUD endpoints, validation and minimum-stock checks; documented APIs with Swagger/OpenAPI and implemented custom exception handling, DTO-Entity conversion, and activity logging.

ASELSAN

Jun. 2025 – Jul. 2025

Intern

Ankara, Turkey

- Developed a Linux kernel keyboard character-device driver with ring-buffer-based storage, an IOCTL interface, and character-device registration for user-space communication.
- Worked with ISR/IRQ handling, workqueues, kernel debugging interfaces, low-level memory management, spinlock-based synchronization, and producer-consumer data flow in kernel space.

Projects

Smart Municipality Application (TÜBİTAK 2209-A) | Flutter, Node.js, TypeScript, YOLOv8, Firebase

May 2026

- Developed an AI-powered cross-platform application enabling citizens to report urban infrastructure problems through a Flutter mobile client.
- Designed an end-to-end architecture combining on-device YOLOv8 Nano inference via TensorFlow Lite, a Node.js/Express backend, Firebase Authentication, Firestore, JWT-based RBAC, and Zod validation.
- Executed edge inference directly on mobile devices to reduce cloud dependency and support offline, low-latency, privacy-preserving reporting.
- Selected for support under TÜBİTAK's 2209-A University Students Research Projects Support Program.

Chord Analysis in Musical Signals | Python, PyQt5, Librosa, Matplotlib | GitHub

Dec. 2024

- Built a desktop application for chromagram extraction, beat tracking, BPM estimation, and audio-signal visualization; implemented major/minor chord detection using template matching and smoothing.

Ashy's Cursed House – VR Horror Escape Room | Unity, C# | GitHub

Jan. 2025

- Built VR puzzle mechanics, object interactions, and hidden-code challenges in Unity/C#, collaborating on level design, puzzle logic, and code integration for Oculus Rift and HTC Vive.

Technical Skills

Languages: Java, Python, C, JavaScript, TypeScript, SQL, HTML/CSS

Backend: Spring Boot, Node.js/Express, FastAPI, RESTful APIs, JPA/Hibernate, Swagger/OpenAPI

Frontend/Mobile: React Native, Expo, React, Flutter, Vite, Vue.js

Databases/Cloud: PostgreSQL, Supabase, Firebase/Firestore

AI/ML: EfficientNet-V2-S, YOLOv8, TensorFlow Lite, K-Means, SVD, TF-IDF, Logistic Regression, LLM Integration (OpenRouter)

Tools/Systems: Git, GitHub, Linux

Programs & Scholarships

Google Yapay Zeka ve Teknoloji Akademisi

Oct. 2024 – Aug. 2025

Scholar

Turkey

- Completed Coursera-based learning tracks in artificial intelligence, technology, and project management and participated in team-based ideathons focused on real-world problem solving.